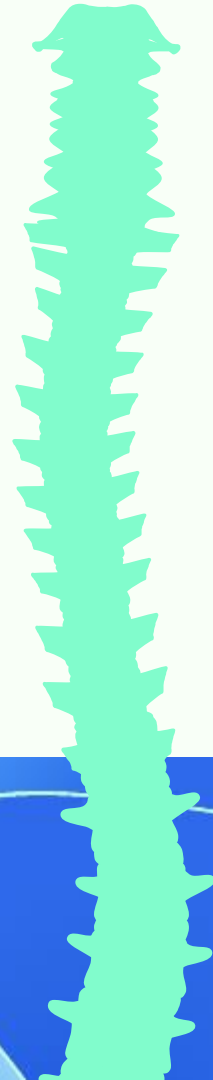




Vertebral Body Tethering (VBT) vs Posterior Spinal Fusion (PSF) for Treatment of Adolescent Idiopathic Scoliosis (AIS): A Systematic Review

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BACKGROUND

Scoliosis is a **complex spinal deformity** that affects **millions of people** worldwide, particularly **adolescents**.

Traditional treatments for scoliosis have relied heavily on posterior spinal fusion (PSF), however, in recent years, a new minimally invasive technique known as vertebral body tethering (VBT) has emerged as a viable alternative

PSF is standard for patients 10 years old or less with curves over 40° until spine maturity, at which point definitive fusion is performed. In contrast, VBT offers an alternative allowing deformity correction and preservation of spinal motion during growth.

This review aims to provide clinicians and surgeons with the reasoning to determine **which treatment suits their patients better** considering different factors.



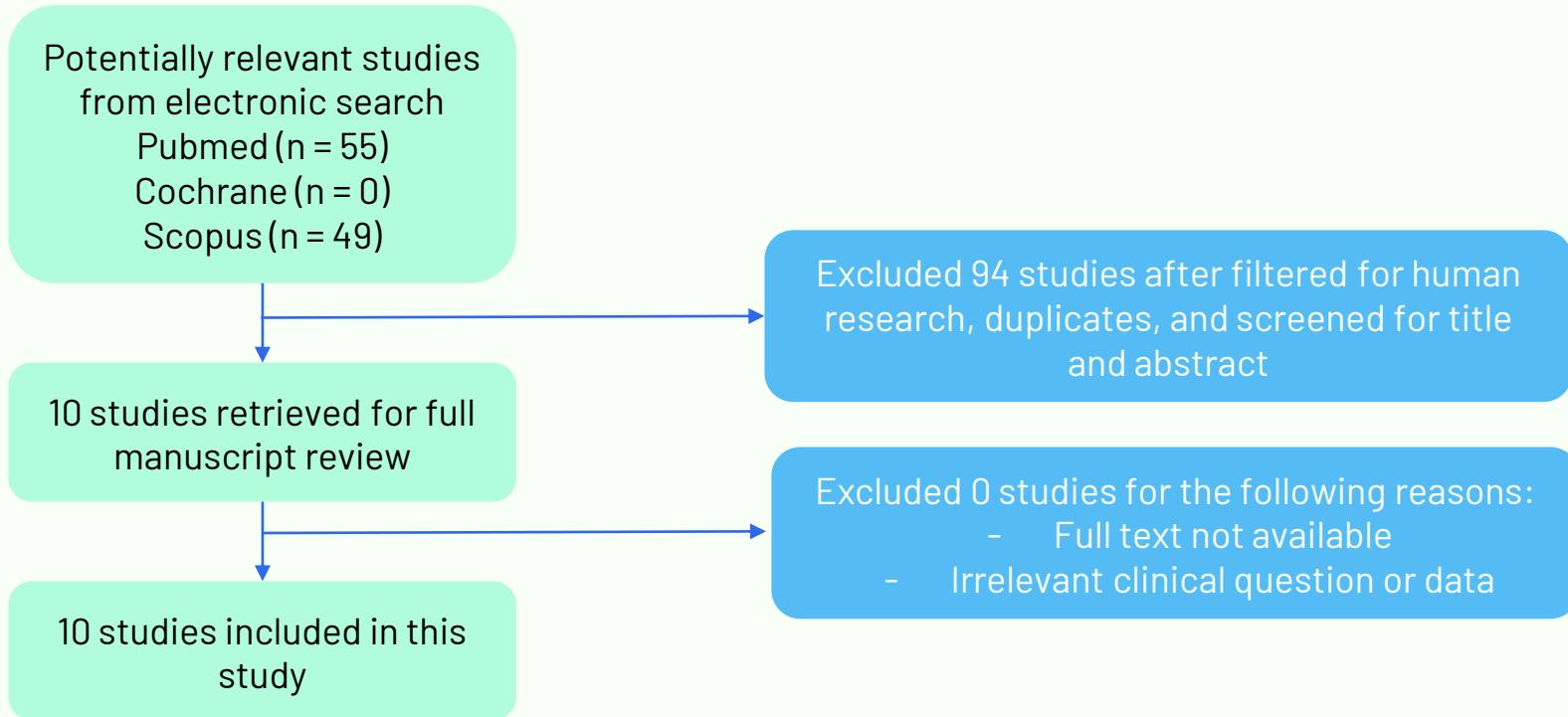
METHODS

Study Design: Systematic Review

Records searching process was done on major medical databases:



METHODS



RESULTS

Study Characteristics

- **Mean follow up** varies from none to 40 months follow up
- **Four prospective studies, six retrospective studies**

Surgery Method

- Consist of **4 comparative studies**, either comparing **VBT and PSF**, **VBT and PI**, or **Harrington instrumentation and posterior spondylodesis**

Safety Aspect

- **Seven studies** reported **no significant adverse events or complication**
 - **Baker et al.** → Mixed results with four revision surgeries
 - **Baroncini et al** → Four complication within first 25 operated patients
- 

RESULTS

Length of Stay



☐ Canbolat et al → overall longer ICU and hospital **stay in VBT**

☐ Baroncini et al 2019 → **4 days in ICU stay** and **hospital stay for 9 days**

Range of Motion



☐ **Mostly diminished spine mobility**

☐ **Nnadi et al** → ROM in the non-fused segment was higher than apical fused segment

☐ **Pehliyanonglu et al** → superior in VBT group trunk flexor and extensor

RESULTS

Pain



☐ **Canbolat et al** → morphine consumption higher in VBT group; NRS higher in VBT group

☐ **Baroncini et al** → reporting low back pain

Bleeding



☐ **Canbolat et al** → less bleeding in VBT

☐ **Baroncini et al** → More blood loss in the first 12 patient compared to the next 13

☐ **Pehliyanonglu et al** → superior in VBT group trunk flexor and extensor

RESULTS

Functional Outcome

- Helenius, et al → Mean Cobb angle 38 degrees for 2-year follow-up and 45 degrees at 20 year follow-up
- Pehliyanglu et al → Endurance → superior in VBT group
- Mostly higher quality of life except **Canbolat et al → no significant difference in preoperative, post, and 3rd week**

DISCUSSION

VBT is often recommended for **skeletally immature patients** because it allows for continued growth and gradual correction of the spinal curvature.

PSI is typically reserved **for more severe curvatures** due to its ability to provide greater stability and correction.

The length of ICU stay in VBT (Mean $\pm$ SD: 2.07 ± 1.21 days) was longer than PSI (1.29 ± 0.77 days), with **significant P = 0.011**.

The length of hospital stay overall in VBT patients was also found longer (6.36 ± 2.50 days) than in PSI (4.70 ± 1.79 days), with **significant P = 0.032**.

DISCUSSION

PSI patients reported **lower pain scores** compared to VBT patients

VBT group had significantly higher postoperative morphine consumption at 1st, 4th, 8th, 12th, 24th, and 48th hours (**$P < 0.05$**).

In patients treated with VBT, they generally experienced **significantly less bleeding** compared to PSI patients.

VBT has also been shown to offer potential health-related quality of life (HRQoL) benefits over spinal fusion, allowing patients to retain a **greater range of motion after surgery**.

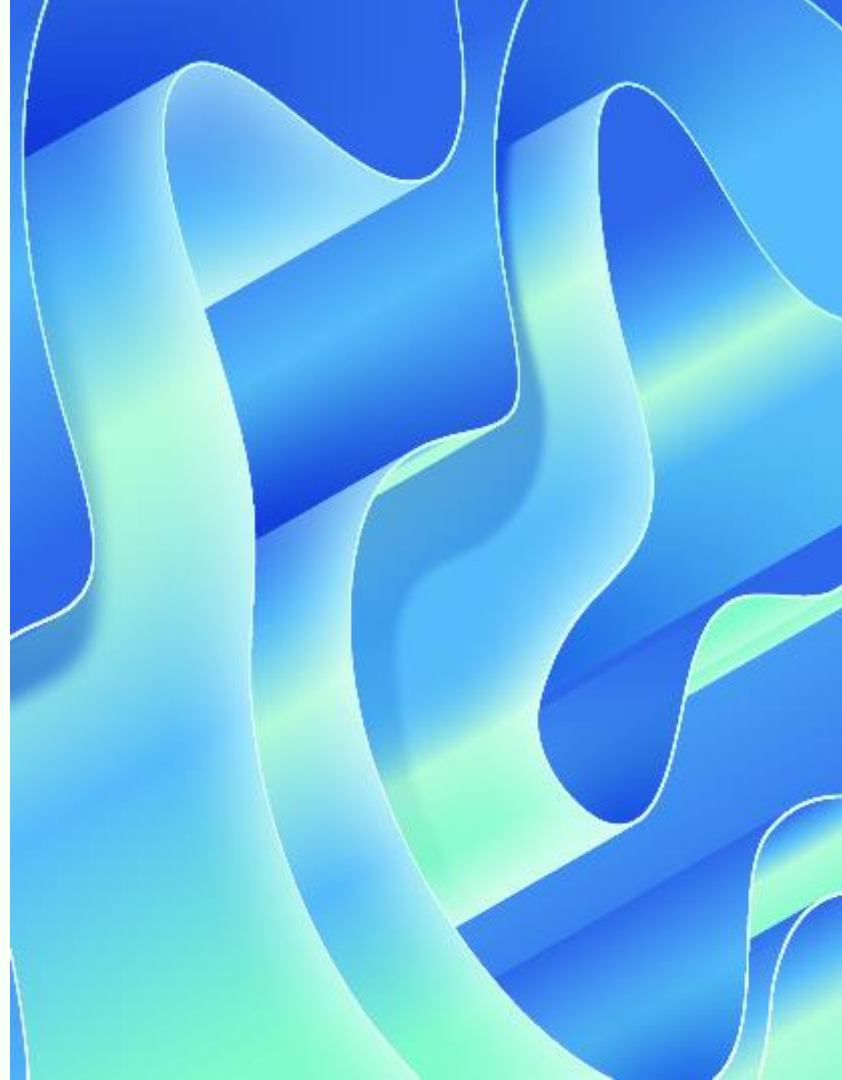
DISCUSSION

The preoperative curve was 45 ± 7 (35–60) degrees, while the final follow-up curve (mean 33 months) was 20 ± 26 (–40 to 25) degrees.

VBT has been associated with the preservation of pulmonary function, making it a **comprehensive approach to AIS management.**

CONCLUSION

- **VBT appears to be the better option in terms of intraoperative bleeding and improving the functional and quality of life** of the patients post-operatively.
- **PSI offers better outcomes in hospital LOS and pain.**
- VBT also offers a less invasive procedure with fewer complications and a lower cost than PSI.
- **Both these surgical methods offer different advantages and disadvantages,** and further research, especially involving human subjects and clinical trials is needed to establish a more robust comparison between these two.



THANK YOU

